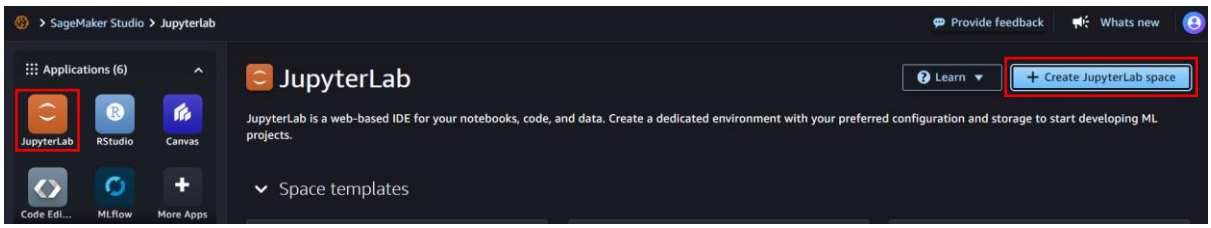
SageMaker Notebooks- Setting up Features

To Begin with the Lab

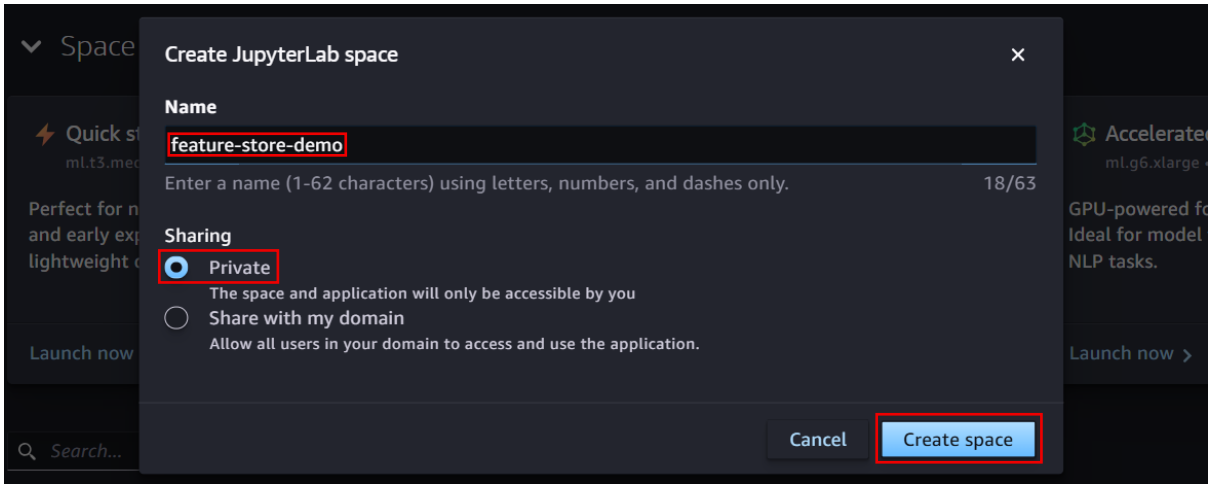
Summary of the Lab

This lab demonstrates how to use a Jupyter Notebook in Amazon SageMaker Studio to work with Feature Store. You create a JupyterLab space, upload and run a notebook, generate sample data, and create feature groups with online and offline storage. Data is ingested and verified via the UI and API. You then query the offline store using Athena. Finally, resources are cleaned up by deleting feature groups and stopping and removing the JupyterLab space.

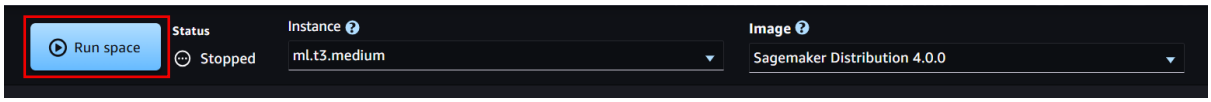
- Lab Objective
    - Use a Jupyter Notebook inside Amazon SageMaker Studio to:
    - Create feature groups
    - Ingest data
    - Retrieve data from online and offline stores
    - Clean up resources
- Open Amazon SageMaker Studio.
- Navigate to JupyterLab.
- Click Create Space.



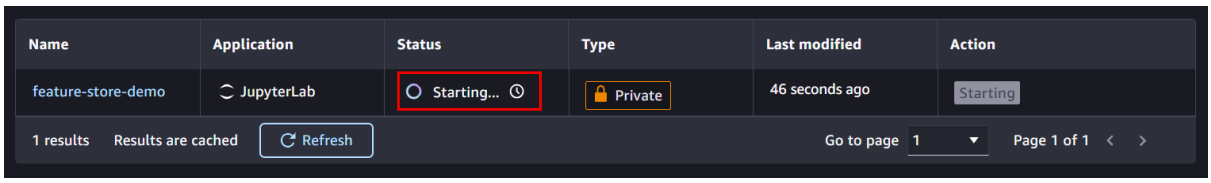
- Enter:
- Name: feature-store-demo
- Type: Private
- Click Create
- Start the space and wait until status is Running



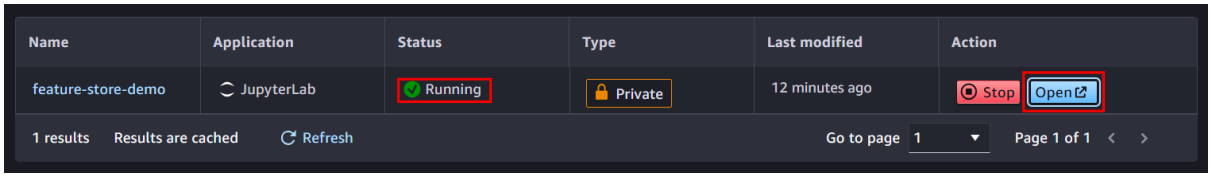
- Click Run Space



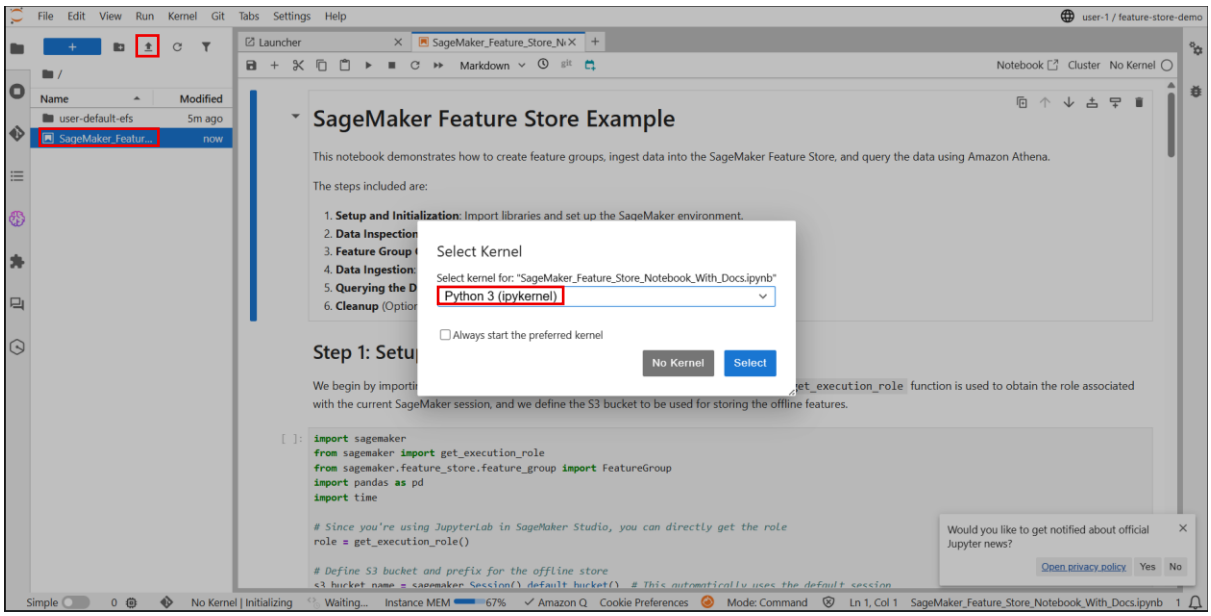
- Click Open JupyterLab once space is ready.



- Click open



- Click Upload Files
    - Select the provided .ipynb notebook
- Upload and open it
    - Choose Python 3 kernel
- Confirm selection



- Run the initial notebook cells to import libraries, configure the environment, and define S3 paths, then generate and display sample customer/order data. Next, create a feature group in Amazon SageMaker Feature Store with a record identifier, event time, and offline store S3 URI, ensuring it is created without errors. Ingest a small dataset into the feature group and verify its presence both in the Feature Store UI and via the GetRecord

API. Finally, query the offline store using Athena from the notebook to retrieve and display stored records.

- This cell creates feature groups in **Amazon SageMaker Feature Store** by defining schema and storing data in both online and offline stores.

```
[3]: # Step 3: Create Feature Groups
customers_feature_group_name = 'customers-feature-group-' + time.strftime('%d-%H-%M-%S', time.gmtime())
orders_feature_group_name = 'orders-feature-group-' + time.strftime('%d-%H-%M-%S', time.gmtime())

customers_feature_group = FeatureGroup(name=customers_feature_group_name)
orders_feature_group = FeatureGroup(name=orders_feature_group_name)

# Generate a proper ISO 8601 string (UTC)
current_time_str = time.strftime('%Y-%m-%dT%H:%M:%S', time.gmtime())

# Assign EventTime
customer_data["EventTime"] = current_time_str
orders_data["EventTime"] = current_time_str

# FIX: Convert datetime column to string (required)
orders_data["order_date"] = orders_data["order_date"].astype(str)

# Load feature definitions
customers_feature_group.load_feature_definitions(data_frame=customer_data)
orders_feature_group.load_feature_definitions(data_frame=orders_data)

# Create Feature Groups
customers_feature_group.create(
    s3_uri=f"s3:///{s3_bucket_name}/{prefix}",
    record_identifier_name='customer_id',
    event_time_feature_name="EventTime",
    role_arn=role,
    enable_online_store=True
)

orders_feature_group.create(
    s3_uri=f"s3:///{s3_bucket_name}/{prefix}",
    record_identifier_name='customer_id',
    event_time_feature_name="EventTime",
    role_arn=role,
    enable_online_store=True
)
```

```
[3]: {'FeatureGroupArn': 'arn:aws:sagemaker:us-east-1:463646775279:feature-group/orders-feature-group-30-18-12-46',
 'ResponseMetadata': {'RequestId': 'ca22d0c2-e674-4a09-a2bf-5e0e970f968d',
  'HTTPStatusCode': 200,
  'HTTPHeaders': {'x-amzn-requestid': 'ca22d0c2-e674-4a09-a2bf-5e0e970f968d',
   'strict-transport-security': 'max-age=47304000; includeSubDomains',
   'x-frame-options': 'DENY',
   'content-security-policy': "frame-ancestors 'none'",
   'cache-control': 'no-cache, no-store, must-revalidate',
   'x-content-type-options': 'nosniff',
   'content-type': 'application/x-amz-json-1.1',
   'content-length': '109',
   'date': 'Thu, 30 Apr 2026 18:12:50 GMT'},
  'RetryAttempts': 3}}
```

- This cell ingests data into the feature groups, making it available for real-time (online) and batch (offline) access.

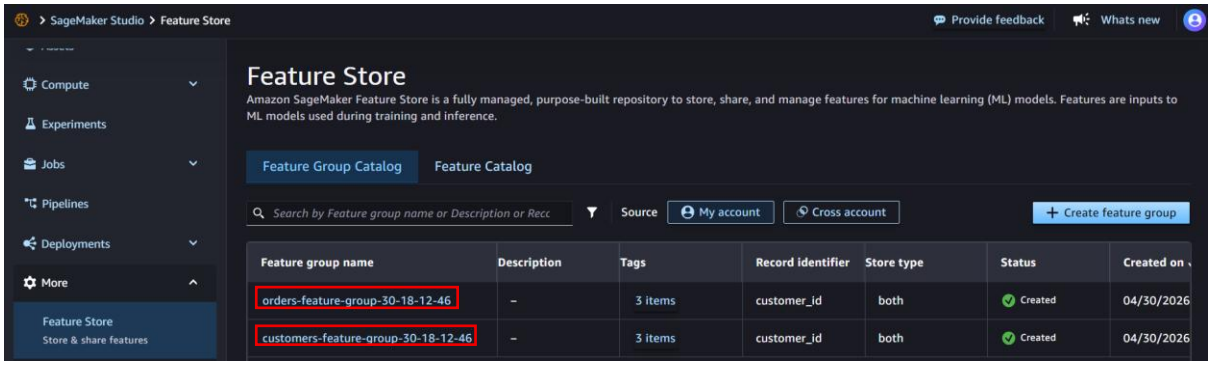
Step 4: Data Ingestion

Once the feature groups are created, we can ingest the customer and order data into the respective feature groups. The data is stored in both the online and offline stores of SageMaker Feature Store.

```
# Wait for feature groups to be created
def check_feature_group_status(feature_group):
    status = feature_group.describe().get("FeatureGroupStatus")
    while status == "Creating":
        print("Waiting for Feature Group to be Created")
        time.sleep(5)
        status = feature_group.describe().get("FeatureGroupStatus")
    print(f"FeatureGroup {feature_group.name} successfully created.")

check_feature_group_status(customers_feature_group)
check_feature_group_status(orders_feature_group)
```

```
Waiting for Feature Group to be Created
Waiting for Feature Group to be Created
FeatureGroup customers-feature-group-30-18-12-46 successfully created.
Waiting for Feature Group to be Created
Waiting for Feature Group to be Created
FeatureGroup orders-feature-group-30-18-12-46 successfully created.
```

- Navigate to you SageMaker → More → Feature Store

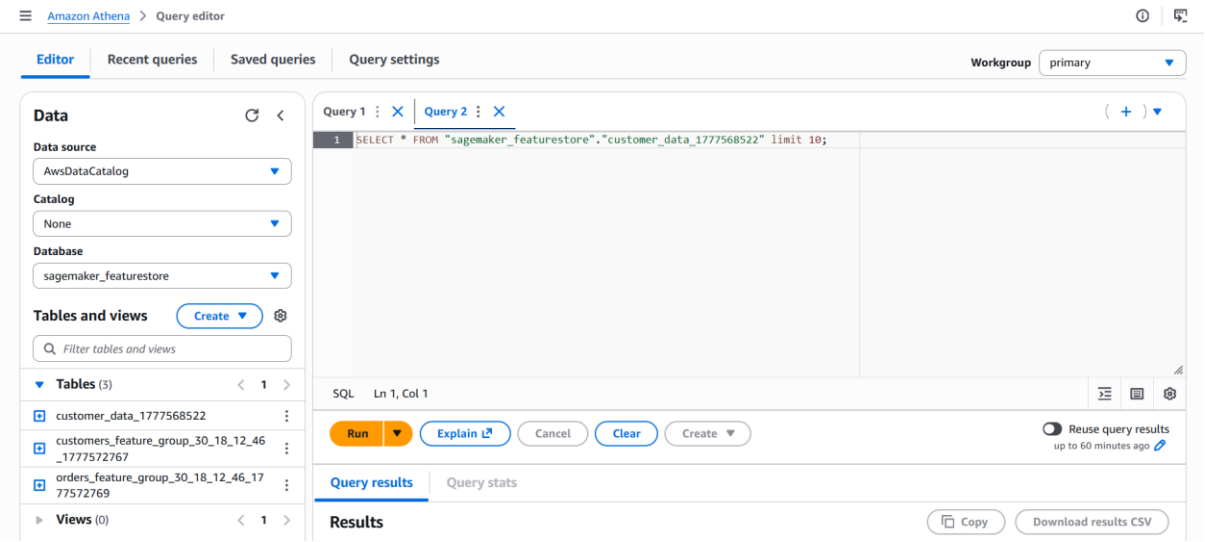


- This cell retrieves a specific record from the online store using the unique identifier, enabling low-latency access for real-time applications.

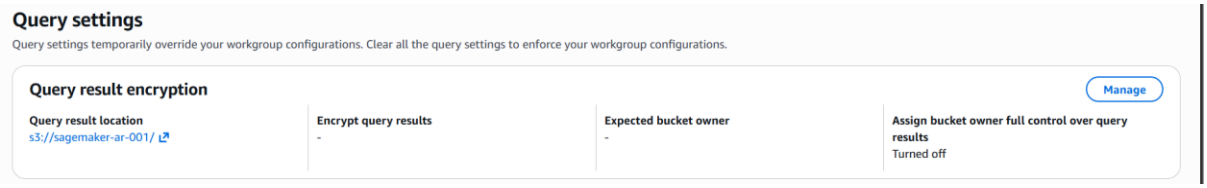
```
[6]: # Step 5: Retrieve Records from the Online Store
import boto3

# Initialize the SageMaker Feature Store runtime client
featurestore_runtime = boto3.client('sagemaker-featurestore-runtime')

# Retrieve a record from the online store
record_id = '1'  # Replace with the actual record ID you want to retrieve
response = featurestore_runtime.get_record(
    FeatureGroupName=customers_feature_group_name,
    RecordIdentifierValueAsString=record_id
)

# Print the record
print("Record retrieved from the online store:")
print(response['Record'])
```

```
Record retrieved from the online store:
[{'FeatureName': 'customer_id', 'ValueAsString': '1'}, {'FeatureName': 'first_name', 'ValueAsString': 'John'}, {'FeatureName': 'last_name', 'ValueAsString': 'Doe'}, {'FeatureName': 'age', 'ValueAsString': '28'}, {'FeatureName': 'account_balance',
'ValueAsString': '1000.5'}, {'FeatureName': 'EventTime', 'ValueAsString': '2026-04-30T18:12:46Z'}]
```

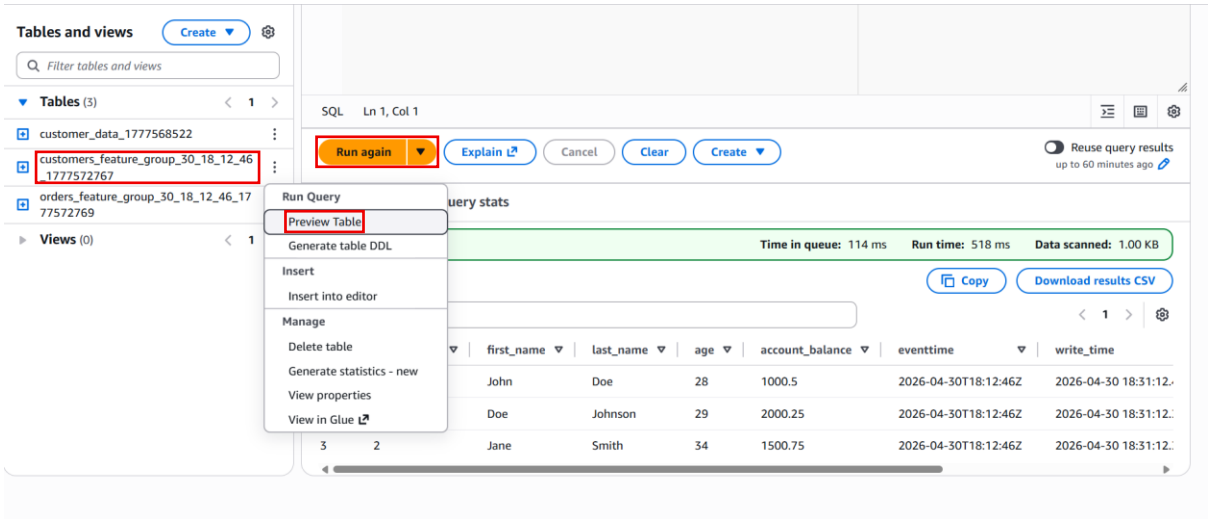
- Open Amazon Athena



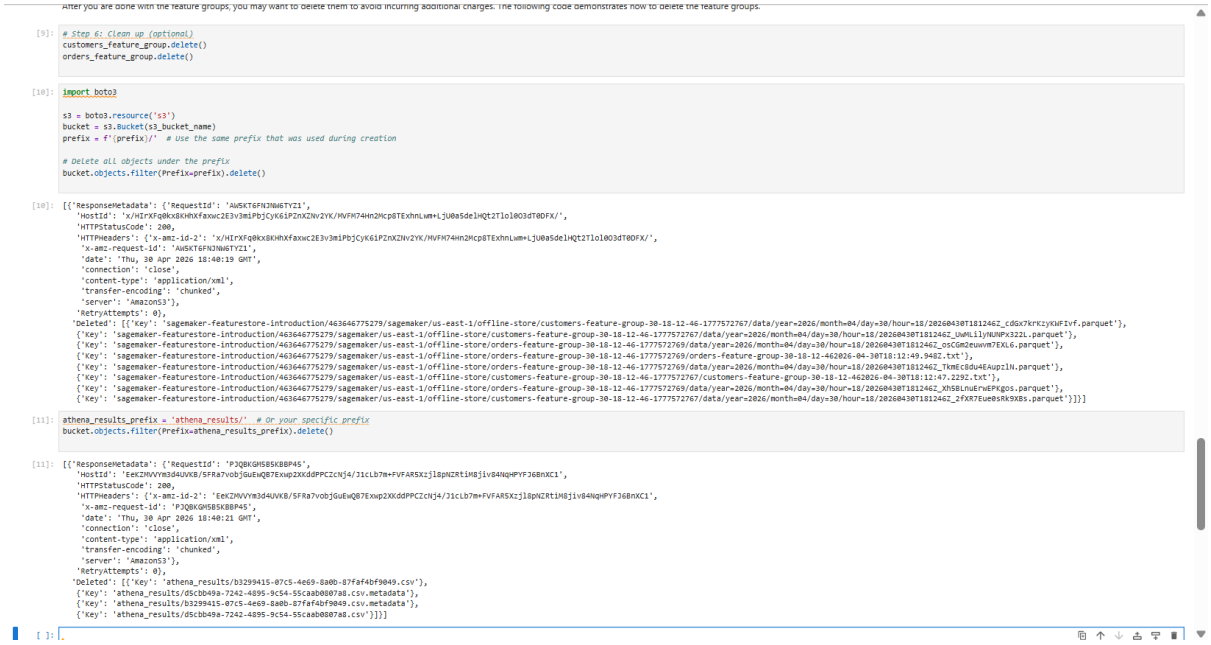
- Set S3 output location (required first time)



- Navigate to:
- Database: SageMaker feature store
- Tables: auto-created feature tables
- Run query or preview table

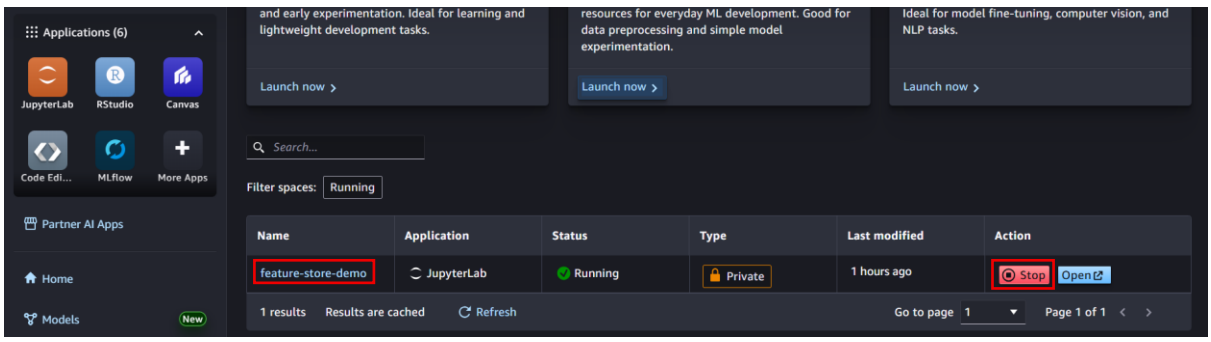
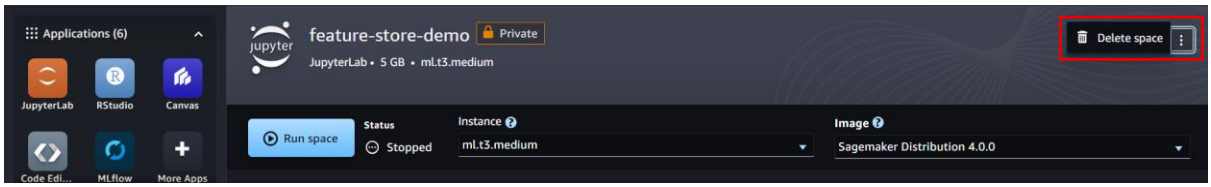


- Delete Feature Groups
- Run cleanup cells in notebook OR
- Delete manually in Feature Store UI



```
[9]: # Step 6: Clean up (optional)
customers_feature_group.delete()
orders_feature_group.delete()

[10]: import boto3

s3 = boto3.resource('s3')
bucket = s3.Bucket(s3_bucket_name)
prefix = f'{prefix}/'  # Use the same prefix that was used during creation

# Delete all objects under the prefix
bucket.objects.filter(Prefix=prefix).delete()

[10]: [{'ResponseMetadata': {'RequestId': 'AW5KT6FNJNW6TYZ1',
    'HostId': 'x/HIrXFq0kx8KHhXfaxwc2E3v3miPbjCyK6iPZnKZNv2YK/MVFM74Hn2Mcp8TExhnLwm+LjU0a5delHQt2Tlol0O3dT0DFX/',
    'HTTPStatusCode': 200,
    'HTTPHeaders': {'x-amz-id-2': 'x/HIrXFq0kx8KHhXfaxwc2E3v3miPbjCyK6iPZnKZNv2YK/MVFM74Hn2Mcp8TExhnLwm+LjU0a5delHQt2Tlol0O3dT0DFX/',
     'x-amz-request-id': 'AW5KT6FNJNW6TYZ1',
     'date': 'Thu, 30 Apr 2026 18:40:19 GMT',
     'connection': 'close',
     'content-type': 'application/xml',
     'transfer-encoding': 'chunked',
     'server': 'AmazonS3'},
    'RetryAttempts': 0},
   'Deleted': [{'Key': 'sagemaker-featurestore-introduction/463646775279/sagemaker/us-east-1/offline-store/customers-feature-group-30-18-12-46-1777572767/data/year=2026/month=04/day=30/hour=18/20260430T181246Z_cdGx7krKzyKWFIvf.parquet'},
    {'Key': 'sagemaker-featurestore-introduction/463646775279/sagemaker/us-east-1/offline-store/customers-feature-group-30-18-12-46-1777572767/data/year=2026/month=04/day=30/hour=18/20260430T181246Z_UwWLilyNUNPx322L.parquet'},
    {'Key': 'sagemaker-featurestore-introduction/463646775279/sagemaker/us-east-1/offline-store/orders-feature-group-30-18-12-46-1777572769/data/year=2026/month=04/day=30/hour=18/20260430T181246Z_osCGM2euwvm7EXL6.parquet'},
    {'Key': 'sagemaker-featurestore-introduction/463646775279/sagemaker/us-east-1/offline-store/orders-feature-group-30-18-12-46-1777572769/orders-feature-group-30-18-12-462026-04-30T18:12:49.948Z.txt'},
    {'Key': 'sagemaker-featurestore-introduction/463646775279/sagemaker/us-east-1/offline-store/orders-feature-group-30-18-12-46-1777572769/data/year=2026/month=04/day=30/hour=18/20260430T181246Z_TkmEc8du4EAupzlN.parquet'},
    {'Key': 'sagemaker-featurestore-introduction/463646775279/sagemaker/us-east-1/offline-store/customers-feature-group-30-18-12-46-1777572767/customers-feature-group-30-18-12-462026-04-30T18:12:47.229Z.txt'},
    {'Key': 'sagemaker-featurestore-introduction/463646775279/sagemaker/us-east-1/offline-store/orders-feature-group-30-18-12-46-1777572769/data/year=2026/month=04/day=30/hour=18/20260430T181246Z_XhSBLnuErwEPKgos.parquet'},
    {'Key': 'sagemaker-featurestore-introduction/463646775279/sagemaker/us-east-1/offline-store/customers-feature-group-30-18-12-46-1777572767/data/year=2026/month=04/day=30/hour=18/20260430T181246Z_2FkR7EuE0sRk9XBs.parquet'}]}]

[11]: athena_results_prefix = 'athena_results/'  # Or your specific prefix
bucket.objects.filter(Prefix=athena_results_prefix).delete()

[11]: [{'ResponseMetadata': {'RequestId': 'P3QBKGM5B5KBBP4S',
    'HostId': 'EeKZMVVYm3d4UVKB/5FRa7vobjGuEwQB7Exwp2XKddPPCZcNj4/J1cLb7m+FVFAR5Xzjl8pNZRtiM8jiv84NqHPYFJ6BnXC1',
    'HTTPStatusCode': 200,
    'HTTPHeaders': {'x-amz-id-2': 'EeKZMVVYm3d4UVKB/5FRa7vobjGuEwQB7Exwp2XKddPPCZcNj4/J1cLb7m+FVFAR5Xzjl8pNZRtiM8jiv84NqHPYFJ6BnXC1',
     'x-amz-request-id': 'P3QBKGM5B5KBBP4S',
     'date': 'Thu, 30 Apr 2026 18:40:21 GMT',
     'connection': 'close',
     'content-type': 'application/xml',
     'transfer-encoding': 'chunked',
     'server': 'AmazonS3'},
    'RetryAttempts': 0},
   'Deleted': [{'Key': 'athena_results/b3299415-07c5-4e69-8a0b-87faf4bf9049.csv'},
    {'Key': 'athena_results/d5cbb49a-7242-4895-9c54-55caab0807a8.csv.metadata'},
    {'Key': 'athena_results/b3299415-07c5-4e69-8a0b-87faf4bf9049.csv.metadata'},
    {'Key': 'athena_results/d5cbb49a-7242-4895-9c54-55caab0807a8.csv'}]}]

[ ]:
```

- Stop JupyterLab Space
- Stop running instance
- Click on Stop First then Delete



- Delete space completely